Text-Based Measures of Cyber Risk
The Economic Impact of Cyber Risk News

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Abstract

We analyze the impact of cyber risk public information on equity returns, at the industrial sector level in the US market. To this aim we build a text-based measure of cyber risk news published by five large diffusion newspapers and we study the dynamic effect of shocks in this measure on weekly equity returns. The analysis shows that the effect propagates in the system and reaches the highest impact one month later. At this time horizon, the effect is positive for all the industrial sectors, including those that are more exposed to cyber risk and negatively affected in the first week. At the one-month horizon, the news seems to trigger a aresilience effect that dominates the increase in expected loss, even though this is statistically significant only for a subset of the industrial sectors, and strongly significant for the firms that provide cyber security.

1 Introduction

The new developments of AI and digitalization of production processes and other corporate activities have brought up the issue of cyber risk to the forefront of the discussion on corporate and financial risk management. From the point of view of corporate risk the impacts of cyber risk can be analyzed at the micro economic level by a measure of the probability and severity of a cyber attack on the operations of the firm. Gauging this kind of risk is particularly difficult because of the lack of data. The data required should cover at least the disclosure of the attack event and information on the level of exposure of each firm in terms of operations digitalization. Unfortunately reporting of attacks is not mandatory in many countries, and even in the US it has become a disclosure requirement only since 2023. The alternative is to evaluate the response of the valuation of corporate liabilities, namely equity and debt, after events which have been publicly known. Companies that are more exposed to cyber risk should record a decrease in their equity values, or an increase

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in their credit spreads after well known episodes of attacks. These effects should be signals of a decrease in expected future dividends and increase in credit spreads. Until detailed firm-wise information will not be available, and made credible by a regulation of mandatory disclosure, relying on public information is the only way to produce a quantitative measure of the effect of a cyber attack.

Early research has approached the problem by means of standard event studies around publicly known cyber attack events. Campbell et al. (2003) extract events from newspapers in the period from 1995 to 2000, collecting evidence of 84 attacks then restricted to 43 events, associated to 38 firms. The analysis does not find a statistical impact on equity, unless the sample is restricted to the events of "security breaches". Cavusoglu et al. (2004) also collect a set of 225 information breach events referred to the period 1996-2001. They found that firms, on average, lost 2.1% of their value two days after the breach, for an overall value of $1.65 billions. They also looked at firms providing information security services that are expected to gain from information breaches to other firms, and found that they increased their value by $1.06 billions.

Our paper shares with this literature the fact that the source of information are the newspaper, and in this sense extends the analysis to a more recent sample, ranging from January 2014 to November 2022. Differently from that literature, we use text-based information to construct an "attention index" of cyber risk, in the same spirit of the literature that applies this kind of analysis to the so called "emerging risks", like climate change (Engle et al. (2020), Ardia et al. (2023)) and geopolitical risk (Caldara and Iacoviello (2022)). In this approach, the focus is on the systemic content of cyber risk, which may have a more general effect on the market as a whole, beyond the individual firms attacked. For example, some firms may not be directly exposed to cyber risk, but nevertheless hit by an event because some other firm in the supply chain is exposed.

More recent research has applied advanced text-based analysis to cyber risk. Florakis et al. (2023) use text based analysis on balance sheet to build a risk-measure of exposure to cyber risk, and find that firms exposed to cyber risk earn a risk premium of 8.3%. Jamilov et al. (2023) also apply text-based analysis to a data set spanning 13 000 firms from 85 countries in from 2002 to 2021. In a bottom-up approach, they model both the effect at the individual level and at the sector level. They also find evidence that cyber risk is priced in the equity and option market. At the global level, the paper estimates a cost of cyber risk around $200 billions.

Differently from this approach, we adopt a top-down strategy checking whether public newspaper information has a significant impact on the market valuation of industrial sectors. Our goal is twofold. First, we want to construct an attention index which could explain the relevance of cyber risk for the economy as a whole. Second, we want to evaluate the dynamics impact of shocks in the attention index on equity returns, and check the sign and statistical signifance of the effect in different sectors and on different horizons.

While the idea aligns with approaches used in climate change, we will show that cyber risk hides peculiarities that make the problem particu-

larly involved. The main difference is the heterogeneity of potential losses triggered by different kinds of cyber-risk, which is much more complex than the distinction between physical risk and transition risk in climate change. So, firms that are affected by some kind of cyber-risk may take advantage of others, and this may change as new typologies of cyber risk appear. In fact, we find that an increase in cyber risk coverage of cyber risk has a positive impact on equity sector indexes returns with a pick at the horizon of one month. As in Cavusoglu et al. (2004), we find that this effect is particularly significant in the information technology sector. However, the sign of the effect at one month distance is positive for all the industrial sectors, even for sectors that are negatively affected (albeit not statistically significant) in the first week after the shock.

The plan of the paper is as follows: Section 2 outlines the theoretic approach for which we would expect that attention in the general public about cyber risk could have an impact on market prices of equity. Section 3 describes the data used in the analysis, including text-based data extracted from newspapers, information from 8-K filings, and market price data Section 4 presents the statistical analysis and discusses the results. Finally, Section 5 concludes.

2 Equity Valuation with Cyber Risk

Here we introduce cyber risk in a standard valuation model:

$$V_t^{(i)} = E\left(\frac{M_{t+1}}{M_t}\left(X_{t+1} - \pi_{t+1} l^{(i)} - z_i\right)\right) \tag{1}$$

where $V_t^{(i)}$ is the equity value at time t of company i, X_{t+1} is the future cash flow and M_t is the stochastic discount factor (SDF). Cyber risk is represented by the conditional probability (intensity of a cyber attack) π_{t+1} at time $t+1$ and the exposure of the firm to cyber risk. In the world famous model of the information security economics, due to Gordon and Loeb (2002) the exposure is represented as

$$l^{(i)} = S\left(\nu^{(i)}, z^{(i)}\right) L^{(i)}. \tag{2}$$

$S\left(\nu^{(i)}, z^{(i)}\right)$ is called "breach probability function" and represents the vulnerability of firm i as a function of intrinsic vulnerability $\nu^{(i)}$ and the investment in security $z^{(i)}$. $L^{(i)}$ is the loss suffered by the company if the cyber attack succeeds.

Two important observations are in order. First, the probability of a cyber attack is independent of the probability of success of it. This is the assumption made in the Gordon and Loeb model. Dropping the hypothesis would only complicate the representation of the vulnerability function, which would make $\nu^{(i)}$ a function of the probability of attack (namely the partial derivative of a copula function). Second, and more important, the probability of attack is not firm specific, while the severity of the attack is. Unfortunately, we cannot observe the security investment policies and the vulnerabilities of the companies. We only know that,

given a common probability of a cyber attack, the equity value of a firm that is more vulnerable than the others will drop more as a result of an increase of cyber risk.

A question is whether some firms can in fact gain value as a result of an increase of cyber risk. Formally, we could modify our basic model as

$$V_t^{(i)} = E\left(\frac{M_{t+1}}{M_t}\left(X_{t+1} + x^{(i)}(\pi_{t+1}) - \pi_{t+1}l^{(i)}\right)\right) \tag{3}$$

where $x^{(i)}$ is the cash flow increase that the company can possibly obtain if the probability of a cyber attack increases. This is not new in the literature: as a matter of fact, Cavusoglu et al. (2004) find that there is a transfer of wealth from IT companies exposed to cyber attacks and firms specialized in providing security against. However, the evidence in favor of gains from cyber risk may extend beyond this kind of firms: any firm that invests in more effective cyber security techniques than the competitors may gain market share when cyber attacks curb the competition.

The asset pricing model can be solved in the standard way

$$\begin{aligned} V_t^{(i)} = {} & B(t,t+1)\mathbb{E}(X_{t+1}) + \mathrm{Cov}\left(\frac{M_{t+1}}{M_t}, X_{t+1}\right) \\ & + B(t,t+1)\left(x^{(i)}(\pi_{t+1}) - \pi_{t+1}l^{(i)}\right) \end{aligned} \tag{4}$$

where $B(t,t+1)$ is the risk free discount factor between time t and $t+1$ and we assume that π_{t+1} and the SDF M_{t+1}.

The focus of our work is on the characterization of π_{t+1}. First, let say what π_{t+1} is. We called it probability of an attack, but it has to be understood as a risk measure, or the probability that the market, or a representative investor, attributes to a cyber attack and the effects it may have on a company in which he wants to invest. Actually, if we abandon the expected utility theory on which the standard asset pricing model is usually based, π_{t+1} may also be interpreted as a "weight function" in the setting of prospect theory proposed by Kahneman and Tversky (1979). In this behavioral framework, investors would evaluate the impact of cyber risk on a company as a lottery in which the weight function is a distorted subjective probability and π_{t+1} and the loss $l^{(i)}$ is what Kahneman and Tversky call "the value function".

Either a probability or a "weight function", π_{t+1} is unobserved. We can only conceive proxies for this measure. A conjecture is that the weight that is assigned to cyber risk could be higher when the issue is much debated on the media. For this reason the probability of risks for which we do not have much of a historical evidence and experience the standard practice is to resort to text-based measures. If we denote o_t such measure it is than reasonable to conceive a function $\pi_{t+1} = \pi(o_t)$. The implementation of this implies the solution of a series of issues. The first is the set of problems about the construction and the statistical properties of o_t. The second is the relationship between o_t and the distorted probability π. We address some of these issues in the empirical analysis below.

3 The Data

We use three sources of data, two of which are text-based. One text-based measure is aimed at identifying cyber-risk as a possyble systemic risk affecting the market as a whole or some industry sectors: this is obtained by constructing a "Ciber News Attention Index" from the analysis of a representative set of newspapers. The other text-based measure is firm-specific, it is based on 8K-form disclosures from each firm attained by a cyber-attack. The third source of data are common stock market prices for a set of firms representative main GICS sectors, of course including those that reported an attack.

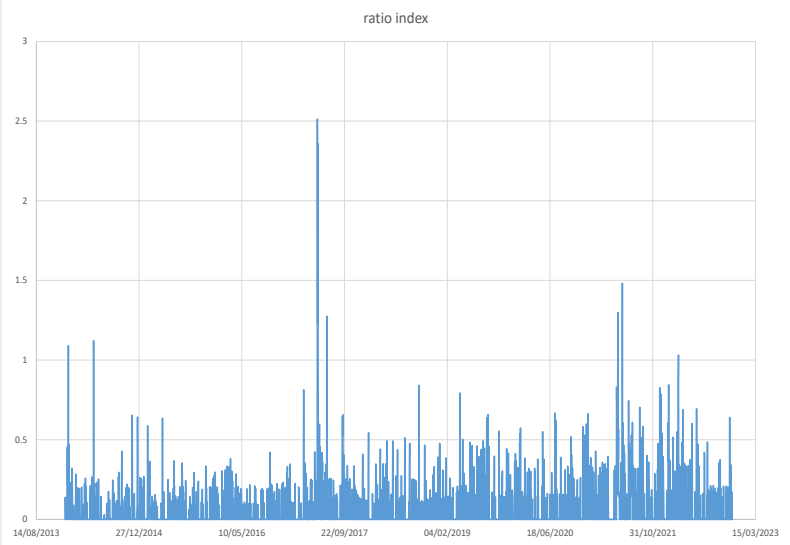
3.1 The Cyber News Attention Index

We define "Cyber News Attention Index" (CNAI) an index of attention of the media and public opinion about cyber risk. As it happens in other applications, the index was built by searching articles addressing cyber risk issues by leveraging textual analysis and unsupervised filtering techniques applied to a corpus of news articles collected from five major newspapers: *Financial Times*, *Los Angeles Times*, *New York Times*, *USA Today*, and *Wall Street Journal*. The dataset comprises articles published over a ten-year period, from January 2014 to December 2023, providing a rich temporal horizon for capturing long-term trends and shifts in media attention. The pipeline ingests raw data from PostgreSQL tables corresponding to each news outlet, processes articles in batches, and assigns to each article a cyber risk relevance score. This score is derived from the relative frequency of a curated set of domain-specific keywords (e.g., "malware", "phishing", "data breach"), and is then normalized to a 0–10 scale to ensure comparability across articles and sources. Textual preprocessing includes lowercasing, tokenization, removal of punctuation and stopwords, as well as exact duplication filtering, yielding a cleaned and deduplicated corpus that is stored in a dedicated relational structure. On top of this enriched dataset, we compute daily, weekly, and monthly indices by measuring the proportion of cybersecurity-relevant articles to the total volume of news published within each time window. The resulting proportion-based indicators serve as interpretable proxies for the evolution of public and media attention to cybersecurity issues. As described in more detail below, to assess its external validity, the behavior of the index was compared to public interest data from Google Trends, showing a strong alignment in the identification of major cyber attack episodes, thereby confirming the index's ability to effectively capture salient cybersecurity events as reported in publicly available news sources. In designing the index, we have deliberately prioritized simplicity and transparency over more sophisticated linguistic or semantic models. This choice ensures that the methodology remains easily auditable and interpretable, facilitating validation and reproducibility across different use cases and datasets. While more advanced techniques such as named entity recognition, topic modeling, or transformer-based embeddings could potentially enhance precision, the current design allows for an intuitive link between raw inputs and index values, which is particularly valuable for policy and

regulatory applications that demand explainability.

Figure 1 presents the time series of the Cyber News Attention Index (CNAI) from August 2013 to March 2023. The index exhibits substantial time variation, characterized by multiple sharp spikes and clusters of elevated attention. These peaks correspond to periods of heightened media coverage of major cybersecurity incidents. The time series appears to be stationary around a low mean with transitory spikes, suggesting that public attention to cyber risk is episodic and event-driven rather than persistent.

Figure 1: Cyber News Attention Index



Notably, there are dense clusters of activity during the mid-2017 period and again in 2021–2022, aligning with globally impactful incidents such as the WannaCry ransomware outbreak in May 2017 (Mohurle and Patil, 2017), the NotPetya malware attack in June 2017 (Greenberg, 2019; Gartzke and Lindsay, 2018), and the Colonial Pipeline disruption in May 2021 (Wilson and Zehra, 2022). These events represent systemic cyber shocks that attracted widespread media attention and significantly impacted economic sectors, making them ideal anchors for the validation of a text-based attention index.

In order to validate and identify these periods of high attention we apply the following procedure:

- we compare the index with measures, extracted from Google Trends,

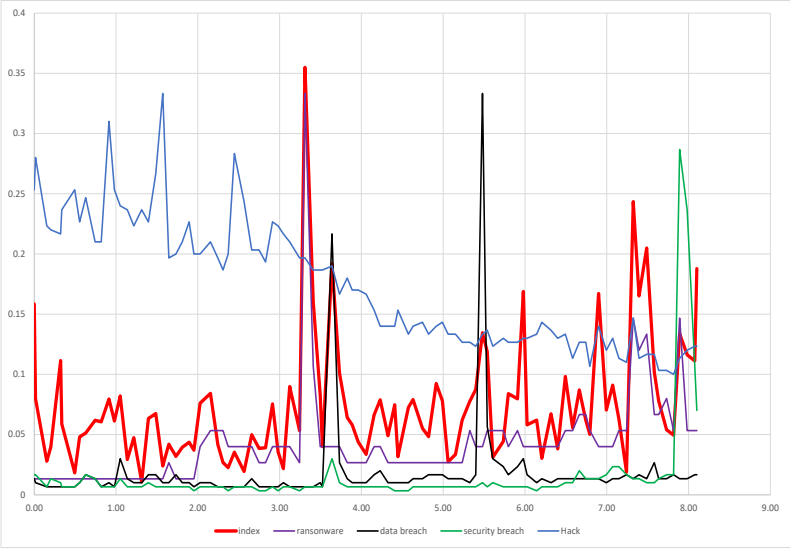
of people attention to a subset of special cyber risk topics, namely:

  - ransomware
  - data breach
  - security breach
  - phishing
  - denial of service
  - hacker attack;

- we identify "months of attention", selecting those in which the frequency of cyber related news, out of the overall number of news, is above a given threshold;
- we use ChatGPT to identify the main events explaining the "months of attention".

The following Figure 2 compares the monthly evolution of the CNAI with normalized Google Trends data for selected cyber risk related keywords.

Figure 2: CNAI and Google Trend Measures



The co-movement between the CNAI and these public interest measures provides external validation of the index. Several prominent peaks in the CNAI correspond closely with those in Google Trends, indicating that the index effectively captures real-time shifts in societal concern about cybersecurity issues. However, the relationship is not perfectly synchronized

in all cases, which may reflect differences in the lag between media reporting and public search behavior. The overlap in spikes further supports the interpretation of CNAI as a proxy for public sentiment shocks related to cyber risk, thereby justifying its use in subsequent empirical asset pricing analysis.

The next step of our work will consist of evaluating the impact of events and innovations in the CNAI index on the prospects of the market, and specific sectors that for some reasons could be particularly exposed to cyber risk.

Table 1: Months of Attention

Date	Cyber Security Event Description
January 2014	Yahoo confirmed that a massive data breach exposed the login credentials of up to 3 billion accounts from previous years.
May 2014	eBay revealed a data breach that affected 145 million users, exposing their encrypted passwords.
May 2017	WannaCry Ransomware: A global ransomware attack hit more than 200,000 computers across 150 countries, disrupting many industries.
June 2017	NotPetya Attack: A sophisticated malware targeted Ukraine but spread globally, causing billions in damages.
September 2017	Equifax Breach: Personal data of 143 million people was compromised in one of the largest breaches in history.
October 2017	Bad Rabbit Ransomware: A variant of ransomware hit several European countries, similar to the NotPetya attack.
July 2019	Capital One Breach: Personal data of over 100 million customers in the U.S. and Canada was compromised.
January 2020	Travelex Ransomware Attack: The UK-based currency exchange was crippled by a ransomware attack, impacting services globally.
December 2020	SolarWinds Hack: A massive cyber espionage campaign affected multiple U.S. government agencies.
May 2021	Colonial Pipeline Ransomware: A ransomware attack caused major fuel supply disruptions across the Eastern U.S.
June 2021	JBS Foods Attack: A ransomware attack targeted the world's largest meat processor, leading to a temporary shutdown of operations.
July 2021	Kaseya VSA Attack: A supply chain ransomware attack crippled businesses using Kaseya's IT management software.
December 2021	Log4j Vulnerability: A critical vulnerability in Log4j was disclosed, affecting countless systems worldwide.
February 2022	Conti Ransomware Gang: The Conti ransomware group threatened retaliation against nations intervening in the Russia-Ukraine conflict.
March 2022	Russian Cyberattacks on Ukraine: Russia launched multiple cyberattacks on Ukrainian infrastructure during the conflict.

3.2 8K-form Data

Having an index of attention mostly based on US newspaper, we had to decide how to build a sample of firms to measure the impact of the index. Hopefully, the sample should include firms that are exposed to cyber-risk, either as a risk or an opportunity. Unfortunately, unlike what happens to climate change, no such exposure measure exists. In the literature some attempts were made to exploit balance sheet reports, called 10-K forms. Here we decided to exploit a new source that is available, called 8-K form, that reveal whether a firm was subject to a cyber attack. Here we describe this data source more in detail.

In the United States, financial disclosure obligations for publicly traded companies are governed by the U.S. Securities and Exchange Commission (SEC), which enforces a robust regulatory framework to ensure transparency and investor protection. Two of the most critical instruments in this framework are Form 10-K and Form 8-K. The Form 10-K is an annual comprehensive report that provides a detailed account of a company's financial performance and operations. Its structure includes a wide range of sections, such as business descriptions, risk factors, financial data, management's discussion and analysis (MD&A), and governance practices. In particular, Part I of the form addresses the company's business and risk profile, while Part II covers financial disclosures, and Part III details gov-

ernance and executive compensation. This form serves as the primary source of standardized financial information and strategic disclosures for investors and analysts

Conversely, Form 8-K is a current report used to disclose significant, material events that occur between the periodic filings of Forms 10-K and 10-Q. These events may include mergers and acquisitions, leadership changes, financial restatements, or cybersecurity incidents. Structurally, Form 8-K is organized into multiple sections, each addressing specific categories of events, such as "Registrant's Business and Operations" (Section 1), "Financial Information" (Section 2), and "Corporate Governance and Management" (Section 5). Importantly, Form 8-K is characterized by its timeliness: registrants must file it within four business days of determining that a reportable event has occurred, unless an exemption (e.g., national security) applies.

While both forms contribute to corporate transparency, they serve distinct yet complementary functions. Form 10-K provides a backward-looking, annual synthesis of a firm's performance, including risk disclosures and financial statements subject to audit. In contrast, Form 8-K is reactive and event-driven, facilitating rapid dissemination of unforeseen material developments that may affect investment decisions. The integration of cybersecurity disclosure into both forms exemplifies this duality. Following regulatory updates in 2023, the SEC now mandates that companies disclose cyber risk management strategies, governance structures, and threat mitigation processes in Item 1C of Form 10-K, reflecting a structured and holistic view of cyber risk oversight. Simultaneously, firms must use Form 8-K to report "material" cybersecurity incidents under Item 1.05 within strict deadlines, with allowances for voluntary disclosures under Items 7.01 or 8.01 when materiality remains undetermined.

To retrieve, analyze, and store cybersecurity-related information from SEC EDGAR filings, specifically the 8-K forms we realized a number of python scripts leveraging the edgartools python library to access EDGAR filings starting from July 2023, filtering explicitly for form types "8-K" and "8-K/A." The script then meticulously examines these filings for specific sections, notably "Item 1.05" and "Item 8.01," assessing their relevance to cybersecurity incidents. A significant feature of the script is its methodical approach in identifying cybersecurity content within these forms through keyword-based sentence analysis. Filings are marked as cybersecurity-related when the proportion of cyber risk sentences reaches or exceeds 20%. This data, once extracted, is organized into a structured DataFrame containing details such as company name, ticker, Central Index Key (CIK), dates of reports and filings, form type, and a hyperlink to the filing. Eventually, the processed data is systematically uploaded into a PostgreSQL database table designed specifically for cybersecurity risk analysis, ensuring structured storage and facilitating future analytical queries.

To enhance further the analysis of cybersecurity incidents reported in 8-K filings, we use advanced natural language processing capabilities provided by OpenAI. Specifically, we use the GPT-4 model to classify the type of cyberattack and identify pertinent entities involved, categorized into organizations, individuals, and locations. A python script connects

to a PostgreSQL database to fetch relevant filings, extracts the company name, date of the report, filing date, form type, filing impact, item classification, and detailed item text. Each filing's textual content is then processed using the OpenAI API, instructed explicitly to return structured responses in a JSON format with clearly delineated categories. These results are subsequently parsed, and the attack type, along with identified organizations, are extracted. Finally, the script updates the PostgreSQL database by inserting the derived classification for each record. Robust error handling and logging mechanisms are integrated throughout to ensure traceability, particularly for database operations and API interactions.

3.3 Equity Data

We collected common stock price data from LSEG Workspace for a set of 195 firms. The total number of firms used was the result of the union of three criteria:

- Firms that have filed a 8K-form reporting a cyber attack
- Firms included in the NASDAQ 100
- Most representative for each GICS (Global Industry Classification Standard) sector, by dimension and market share, even if listed on the NYSE

Of course, the information technology sector is expected to play a relevant role in the analysis. For this reason, we decided to split the sample of companies in the sector by creating a sub-sector called "security" including the firms that Cavusoglu et al. (2004) identified as those positively affected by cyber-risk events.

For each company we downloaded the time series of closing prices, which were used to compute weekly returns that will represent the variable of attention of this paper.

Table 2 offers an overview of cyberattack incidence across industry sectors, drawing on a sample of 190 firms classified according to the Global Industry Classification Standard (GICS). Of these, 63 firms reported experiencing a cyberattack, as obtained from 8-K forms information, yielding an aggregate incidence rate of approximately 33.2% of our sample. However, this headline figure conceals considerable heterogeneity across sectors, which carries important implications for understanding cyber risk as a differentiated and sector sensitive phenomenon and reflecting differing levels of vulnerability, digital maturity, and perhaps reporting practices.

From the number of attacks per industrial sector, the 8-K data show a clear ranking:

- the financial sector is the most affected, with 17 attacks (27% of the overall attacks since 2023), corresponding to 70% of the firms in our sample;
- a cluster of sectors, namely information technology, industrial, consumer diversified, industrial and health, record between 8 and 11 attacks each.
- communication, energy and consumer staples are the other sectors that have been reached by at least one attack.

Table 2: The Sample

GICS	# Companies	# Attacked	% Attacks
Security	10	0	0.0
Information Technology	50	11	17.46
Communications	13	3	4.76
Utilities	5	0	0.0
Energy	10	2	3.17
Financial	24	17	26.98
Aerospace & Defense	7	0	0.0
Health	20	8	12.70
Consumer Diversified	22	10	15.87
Consumer Staples	8	1	1.59
Industrials	21	11	17.46
Total	190	63	

The financial sector shows the highest number, and rate of reported cyberattacks, with 70.8% of the financial firms in our sample having disclosed at least one incident. This finding is consistent with theoretical expectations and empirical evidence identifying financial institutions as prime targets due to the sensitivity and concentration of high value digital assets, as well as their systemic role in the economy.

In the second cluster of firms, the industrial and consumer diversified exhibit an elevated number of attacks suggesting that firms embedded in complex, often digitally integrated supply chains may face increased exposure to cyber threats. The health care score, another sector undergoing rapid digital transformation, also emphasizes its attractiveness to attackers due to the sensitivity and value of personal health data.

The information technology and security sectors, and to some extent the communication sector, deserve a different argument. On one hand, the degree of digitalization of these sectors would lead us to expect a huge number of attacks. On the other hand, one may also expect that firms of this kind hold plenty of defense and mitigation techniques available, so that the number of successful attacks would be lower. Consistently with the latter argument, no attack at all was recorded among the firms that we classified as security providers.

Finally, the relatively low incidence rates in sectors such as consumer staples and energy may reflect lower levels of digitization, fewer incentives for attackers, or differences in disclosure practices.

Overall, the distribution of reported attacks across sectors reveals substantial variation that reflects both structural industry characteristics and institutional factors affecting disclosure. These findings support the paper's broader argument that cyber risk cannot be considered a uniform shock, but rather a multifaceted and heterogeneous phenomenon. This underscores the need for differentiated empirical strategies and regulatory frameworks that account for sector-specific exposure, reporting norms, and resilience capacities.

4 Statistical Analysis

The goal of our statistical analysis is to gauge if cyber risk has assumed a systemic role, like the other main "emerging risks" that are spreading uncertainty among the markets, with particular worries for the long run. Like climate change and geopolitical risk, cyber risk in the digitalization era has taken on a relevance that it did not have before. Like those risks, there is huge uncertainty about the new forms and the relevance that it could take in the future.

A natural strategy is to apply to systemic cyber risk the same strategy that has been used for climate change and geopolitical risk. Evaluate the impact of our CNAI index on the excess returns of the stocks in our sample. Since our question is whether cyber risk is priced as a systematic risk factor, we will refer to the excess returns of equally weighted indexes of the GICS sectors.

4.1 Vector Auto Regression

VAR models are particularly useful for unfolding the dynamic relationship between time series by allowing each variable to be influenced by its own other lagged values and by those of other variables in the system. To analyze the potential effect of CNAI shocks on the stock market, we estimate a bivariate vector autoregression (VAR) on $Y_t := (R_t, \Delta CNAI_t)$, where R_t is the return of an equally weighted portfolio of equities. Given the process $Y := \{Y_u, u = 1, 2, \ldots, T\}$, a VAR(p) model can be represented as:

$$\Delta R_t = a_{10} + \sum_{i=1}^{p} (a_{1i}\Delta R_{t-i} + b_{1i}\Delta CNAI_{t-i}+) + \varepsilon_{1,t} \tag{5}$$

$$\Delta CNAI_t = a_{20} + \sum_{i=1}^{p} (a_{2i}\Delta R_{t-i} + b_{2i}\Delta CNAI_{t-i}+) + \varepsilon_{2,t} \tag{6}$$

where a_{ki}, b_{ki}, for $k = 1, 2$ and $i = 0, 1, 2, \ldots, p$ are constant coefficients that capture lagged relationships up to order p, $\varepsilon_t := (\varepsilon_{1,t}, \varepsilon_{2,t}) \sim WN(0, \Sigma)$ is a 2×1 vector of white noise residuals with zero mean and variance-covariance matrix Σ.

The analysis is repeated separately with R_t representing either the mean return of the entire sample of stocks under investigation or the mean returns for each sector of the GICS.

In what follows, we report and interpret the results from the return equation only, as our primary interest lies in understanding the impact of cyber risk attention on asset prices. Notably, the estimation of the second equation, where changes in the CNAI index ($\Delta CNAI$) are regressed on past returns, shows no statistically significant effects, thereby ruling out concerns of reverse causality from returns to cyber risk attention.

The results are reported in table 3. We discuss here the main results contained in the table, concerning the sign and the statistical significance of the impact of a shock in the CNAI index on the returns. For the ease of the reader we collect them in the following items:

- as for the immediate effect of the shock to the index, one week, we find that the impact is negative for the financial sector, aerospace and defence, health (the highest impact) and consumer staples; among the sectors recording a positive impact, the highest is the sector providing security, consistently with the results in Cavusoglu et al. (2004). No impact is statistically significant and the effect on the full sample is mildly positive.
- Starting from the second week, the impact becomes positive across all the sectors (with the exception of aerospace and defence that remains mildly negative in the second week only); the effect remains not statistically significant.
- after one month, the impact is positive in all the sectors, and it is the highest in all of them; only at this horizon we have statistical significance for some sector: the most significant is again the security sector, that is the set of firms selected by Cavusoglu et al. (2004). The other sectors are information technology, communications, utilities and consumer staples. The fourth week impact is also statistically significant on the full sample, and even in this case it is the only time horizon at which statistical significance shows up.

The overall message of these findings seems to be that public opinion attention is not particularly attracted by cyber risk news. For sure, it has much less impact than climate change and geopolitical news. Nevertheless, there seems to be an effect that percolates into the market and provides a significant increase of equity returns one month later. The sign of the effect indicates that the signal of resilience prevails over the effect on the expected loss. The way in which this resilience tidal effect rises over one month is a clear signal that some propagation mechanism is at work. A possible explanation could be the presence of momentum traders, as in the model by Hong anf Stein (1999), who may help the gradual diffusion and anmplification of signals, albeit weak. This could be also corroborated by other mechamism, such as the presence of attention-constrained investors that could be attracted by the trading activity (e.g. Barber and Odean, 2008). Both of these explanations appear particularly plausible for weak signals like the ciber risk news appear to be,

Param.	Sec.	IT	Comm	Util.	Enrgy	Fin	Aer& Def	Health	Cons.Div.	Cons.Stp	Ind.	Full
a_{11}	-0.78***	-0.86***	-0.92***	-0.74***	-0.83***	-0.85***	-0.84***	-0.96***	-0.69***	-0.8***	-0.81***	-0.84***
b_{11}	1.79	1.59	0.70	1.62	1.68	-0.58	-0.78	-1.14	0.46	-0.97	0.26	0.39
a_{12}	-0.68***	-0.66***	-0.77***	-0.62***	-0.63***	-0.68***	-0.71***	-0.73***	-0.57***	-0.66***	-0.64***	-0.66***
b_{12}	0.98	2.93	1.79	1.63	1.34	2.59	-0.25	1.40	0.12	2.32	1.06	1.76
a_{13}	-0.51***	-0.45***	-0.56***	-0.54***	-0.41***	-0.51***	-0.62***	-0.51***	-0.53***	-0.47***	-0.43***	-0.46***
b_{13}	3.64	3.09	3.17	3.52 .	1.40	1.32	1.95	2.35	0.79	1.04	0.57	2.16
a_{14}	-0.42***	-0.39***	-0.41***	-0.40***	-0.41***	-0.46***	-0.45***	-0.48***	-0.42***	-0.46***	-0.43***	-0.43***
b_{14}	7.8***	7.31*	4.87*	4.72*	6.30.	5.38.	3.10	5.59.	2.13	6.97**	4.23.	5.76**
a_{15}	-0.36***	-0.26***	-0.28***	-0.21***	-0.18**	-0.26***	-0.31***	-0.28***	-0.32***	-0.26***	-0.23***	-0.26***
b_{15}	1.90	-0.76	-0.71	2.96	0.49	0.37	0.52	-1.31	0.08	0.44	-1.66	-0.20
a_{16}	-0.17***	-0.16***	-0.17***	-0.16***	-0.12**	-0.22***	-0.21***	-0.20***	-0.17***	-0.24***	-0.17***	-0.19***
b_{16}	2.86	1.73	1.47	1.65	2.17	0.56	1.94	0.12	-0.02	1.60	0.27	1.42
a_{10}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Adj. R^2 (%)	40.48	44.82	46.47	36.34	43.99	44.47	41.87	52.09	33.10	44.33	43.07	44.48

Table 3: VAR results for the full sample and by GICS sectors: focus on the return equation

5 Conclusions

This paper develops a novel text-based approach to assess the economic relevance of cyber risk news, focusing on its perceived intensity in public discourse and its pricing impact in equity markets. We construct a Cyber News Attention Index (CNAI) from major newspapers and explore its dynamic relationship with market returns through a VAR analysis.

Our findings indicate that cyber risk attention, as captured by the CNAI, provides a signal, that, albeit weak, has a clear impact on the market. First, we find that the effect on the first week return, albeit not statistically significant, is negative for the financial, aero-defence, health and consumer staple sectors, which are typically considered the most exposed. Second, in the following weeks a a general positive impact prevails on all the sectors, reaching the highest effects one month after the news. At this horizon, the effect is highly significant for a set of firms that a previous analysis carried out by Cavusoglu et al. (2004) identifies as "security providers". It is also significant for the information technology sector as a whole, the communication, utility and consumer stable sectors. Overall, we find clear evidence of a resilience effect originated from the attention to the cyber news, that prevails on the perception and worries of cyber risk exposure, and that reaches its highest effect one month after the news. The result holds for all the sectors, no matter whether they are exposed to losses from cyber risk, or are expected to gain from it. In this sense this is different from the result in Cavusoglu et al. (2004) which finds positive returns only for security providers.

Future work may get deeper in this analysis checking whether within the sectors there exist features that allow to distinguish whether individual firms behave differently from the sector. The one month effect obtained on realized return could also be compared with expectations, measured from the option markets. The attention measure could also be refined through semantic modeling or other techniques, but more to the point it will be important to have enough history of 8-K forms to put together the attention index and the firm-based information. This will be particularly helpful in a context where regulatory frameworks are evolving and attacks are becoming more sophisticated. In this perspective, real time market based signals such as the CNAI will represent valuable tools for both investors and policymakers.

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